

Load Balancing Methods:

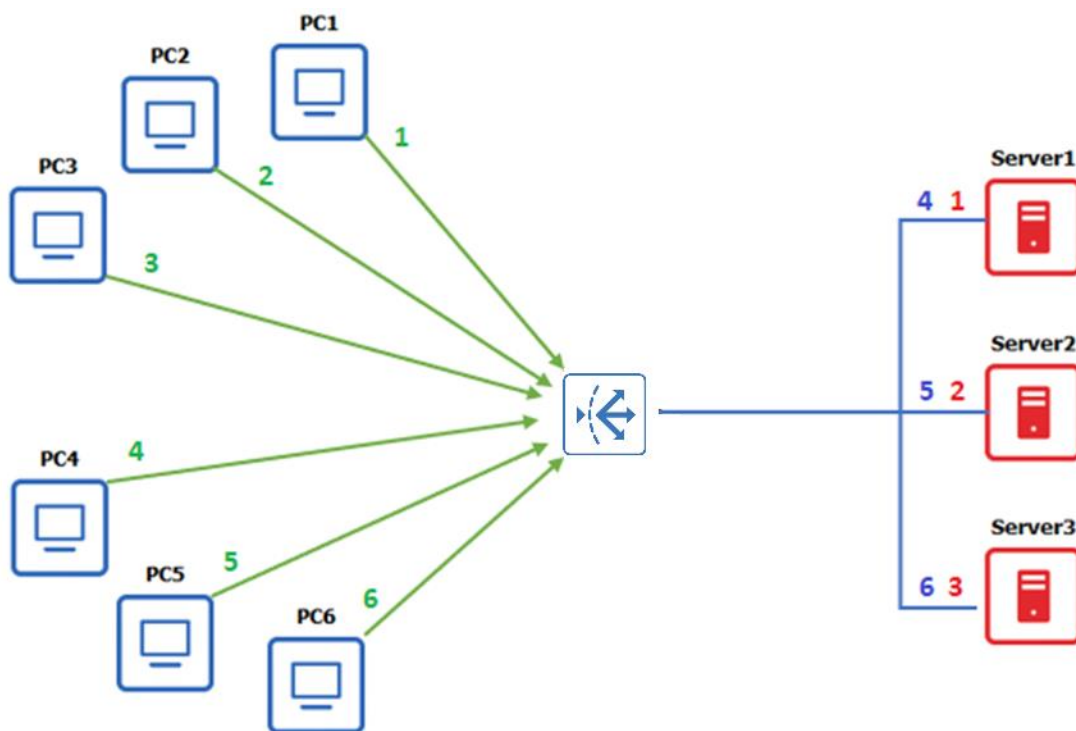
In simple words load balancer is a network device that distributes traffic to other devices. Efficiently distributing incoming the network traffic across a group of backend servers. Performs three interrelated functions; monitoring hosts, acting as proxy & load balancing. Primary intent for load balancing is the same to provide higher availability of applications.

Load Balancing is an important method used to spread network traffic across several servers. This ensures that no single server gets overloaded with too many requests.

FortiADC supports several Load Balancing algorithms to distribute traffic efficiently across backend servers. These algorithms can be configured based on the specific needs of your environment to ensure optimal performance and reliability.

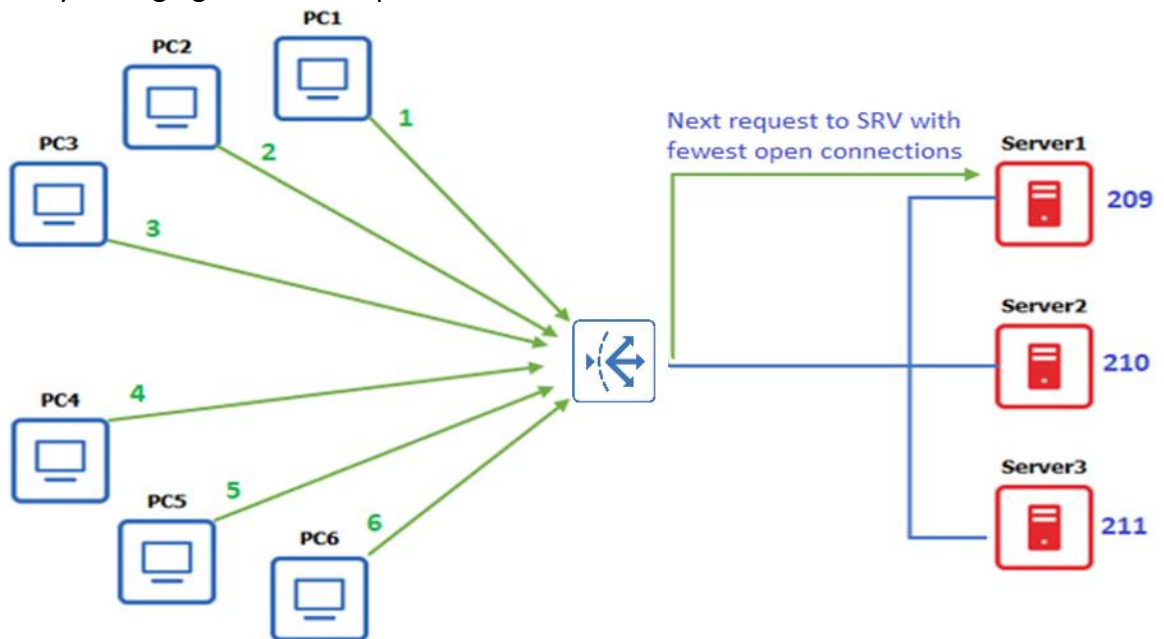
Round Robin:

This algorithm distributes traffic evenly across all servers in a circular order. It's simple and works well when all servers have similar capacity. The traffic load is balanced by rotating through the servers in sequence. This algorithm/Method is designed to equally distributed load across all the pool members. Generally used when we have equal hardware such as Processor, CPU, RAM, and Memory.



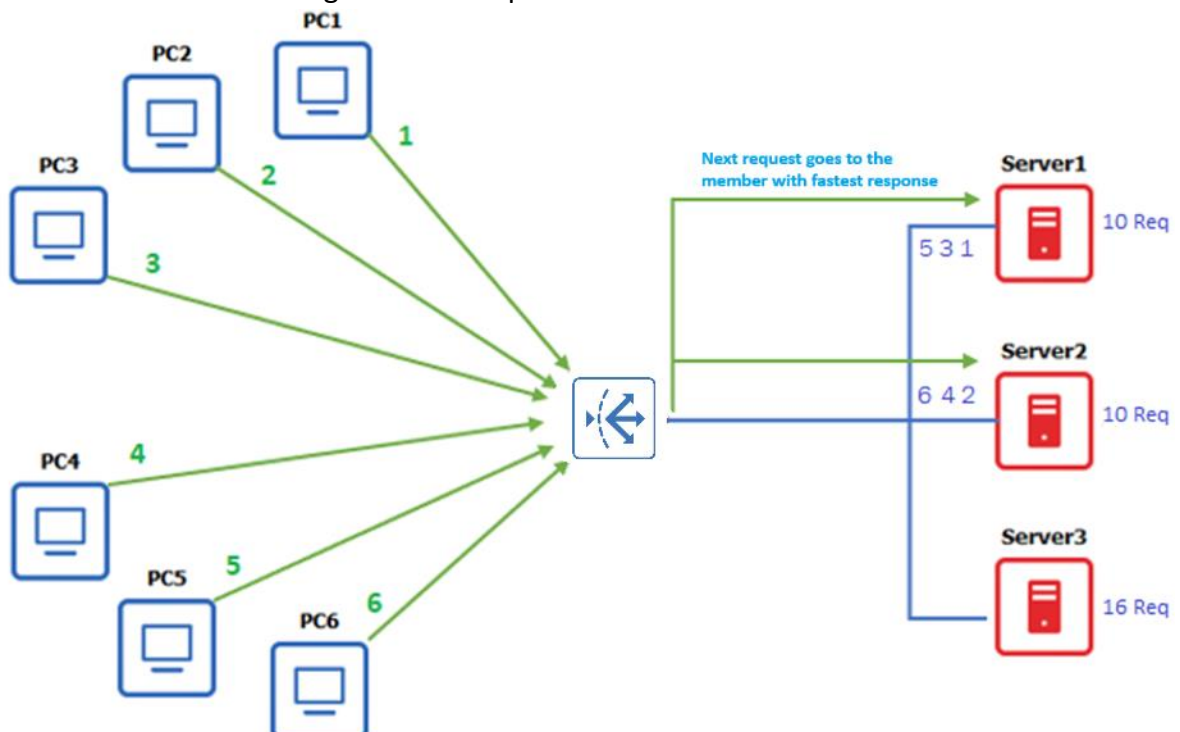
Least Connections:

Traffic is directed to the server with the fewest active connections. This is useful when there are significant differences in the load handled by each server. Distributes connections to server that is currently managing the fewest open connections.



Fastest Response:

Directs traffic to the server that responds the fastest based on the health check tests. This requires continuous monitoring of server response times.

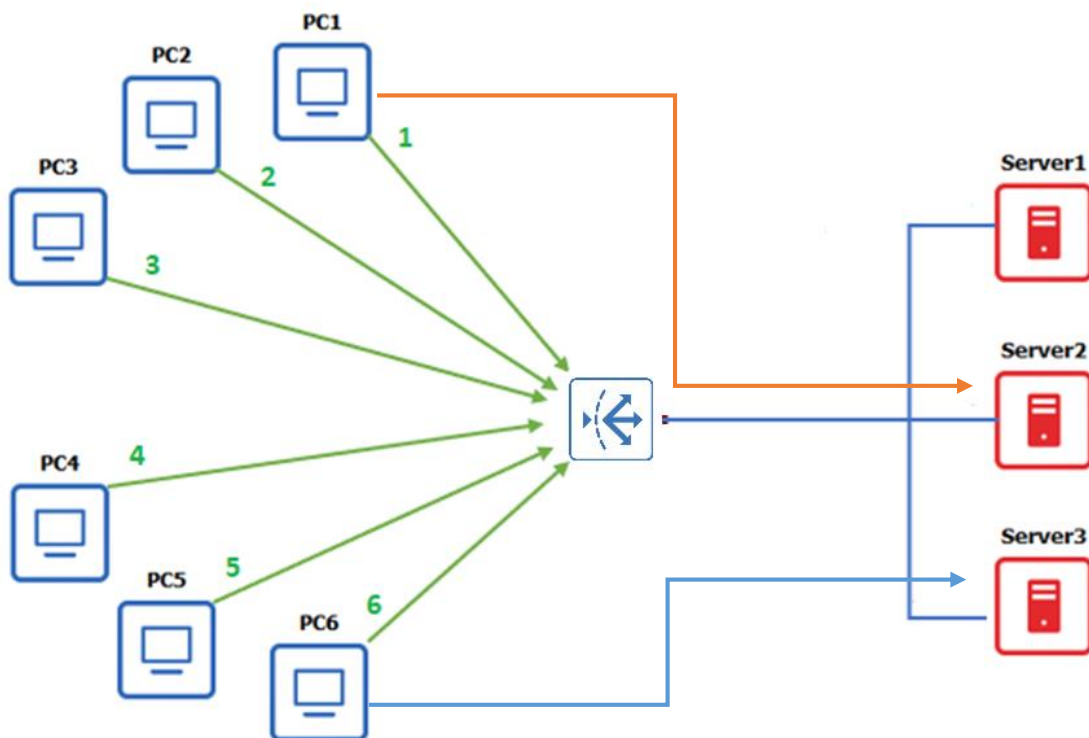


IP Hash:

The client's IP address is used to determine which server receives the request. This ensures that requests from the same client are always sent to the same server, which is useful for session persistence.

Source IP Hash:

Similar to IP Hash, but uses the source IP address of the client to determine the server. Source IP Hash is a load-balancing method that uses the source IP address of the client to calculate a hash. This hash determines which backend server will handle the traffic. It ensures that requests from the same client IP are consistently directed to the same server, providing session persistence without requiring additional mechanisms like cookies. This method is particularly useful in scenarios where maintaining a consistent server-client relationship is essential, such as in applications requiring stateful connections or caching.



Destination IP Hash:

This load-balancing method selects the next hop by applying a hash function to the destination IP address of incoming traffic. The hash determines which backend server will handle the traffic, ensuring consistent routing for the same destination IP. This method is particularly useful for maintaining session persistence without requiring additional tracking mechanisms. However, it is available only when using a Layer 2 Virtual Server on the FortiADC, meaning the ADC operates at the data link layer and forwards packets based on MAC addresses rather than IP headers.

Dynamic Load:

Dynamic Load balancing in FortiADC uses SNMP health checks to monitor backend servers' performance metrics such as CPU, memory, and disk usage. Based on these metrics, weights are assigned to each server, with lower weights indicating less resource usage. Traffic is then dynamically distributed to the servers with the lowest weights, ensuring efficient use of resources and preventing overloading of heavily utilized servers. This approach adapts to real-time changes in server health, maintaining optimal performance and resource balance.

Imagine you have three backend servers monitored by FortiADC using SNMP:

Server A: CPU at 40%, memory at 50%, disk usage at 30%. Assigned weight: 40.

Server B: CPU at 70%, memory at 80%, disk usage at 60%. Assigned weight: 70.

Server C: CPU at 20%, memory at 30%, disk usage at 20%. Assigned weight: 20.

A new request comes to the FortiADC. Based on the assigned weights, Server C (Weight 20) is selected because it has the least resource utilization.

